1.3 Subgroup, Quotient Group

June 12, 2026

* Let G be a topological group and $H \subset G$. Then H is a subgroup of topological group G iff H is a subgroup of group G and H is a closed set in topological space G.

Remark 0.1. H is a topological group equipped with induced topology on G. A subgroup of an abstract group which is a topological group is a topological group. This works as restriction of continuous maps is continuous.

* A normal subgroup N of topological group G if N is a normal subgroup of abstract group G.

Proposition 0.1. Let G be a topological group. Let H be a subgroup of abstract group G.

(i) $\overline{H}$ is a subgroup of top. group G

(ii) If H is normal subgroup of G then $\overline{H}$ is normal.

Proof. (i) Observe $\overline{H} = \bigcap_{V \in \mathcal{V}} HV$ where $\mathcal{V}$ is collection of all neighborhoods of e. As $H \subseteq \overline{H}$ hence $\overline{H} \neq \emptyset$.

For $x, y \in \overline{H}$ we want to show $xy^{-1} \in \overline{H}$.

We know $m : G \times G \to G$ is continuous.

$$(x, y) \mapsto xy^{-1}$$

Observe $m^{-1}(\overline{H})$ is closed, and $H \times H \subseteq m^{-1}(\overline{H})$

Hence $\overline{H} \times \overline{H} = \overline{H \times H} \subseteq m^{-1}(\overline{H})$ and this implies

$$m(\overline{H} \times \overline{H}) \subseteq \overline{H}.$$

Cii) We construct a mapping

$$n : G \times G \to G$$
$$(x, y) \mapsto xyx^{-1}$$

and observe n is a continuous map this implies $n^{-1}(\overline{H})$ is closed and similar to (Ci)

$$H \times H \subseteq n^{-1}(\overline{H})$$

this implies $\overline{H} \times \overline{H} = \overline{H \times H}$ is a subset of $n^{-1}(\overline{H})$. Hence

$$n(\overline{H} \times \overline{H}) \subseteq \overline{H}.$$

$*$ Set of all left cosets set G/H. We have a natural map or projection $\pi : G \to G/H$.

$$x \mapsto xH$$

Remark 0.2. If G is a topological group, we will topologize G/H assuming H is closed. Construct

$$\tau_{G/H} = \{O \subseteq G/H \mid \pi^{-1}(O) \in \tau_G\}$$

i.e. O is open if and only if $\pi^{-1}(O)$ is open in G

which ensures π is a continuous map, moreover a quotient map where π is surjective by construction.

Lemma 0.1. *G/H is a T_1-space, and π is open.*

Proof. Since H is closed then αH is closed by homeomorphism. Observe $G - \alpha H$ is open in G. Let $\tilde{\alpha} = \alpha H \in G/H$. Observe $\pi^{-1}(G/H - \tilde{\alpha}) = G - \alpha H$. Hence $\tilde{\alpha}$ is closed, G/H is T_1.

We want to show π is an open map. Let O be an open map in G/H then $\pi(O)$ is open G/H iff $\pi^{-1}(\pi(O))$ is open in G. Now $OH = \pi^{-1}(\pi(O))$ and $OH = \bigcup Oh$. Hence OH is open. So

$\pi^{-1}(\pi(O))$ is open which implies $\pi(O)$ is open.

Lemma 0.2. *G/H is a T_2 space.*

Proof. Consider $xH \neq yH$ in G/H.

Now $x \in xH$ and $x \in G \setminus yH$ which is open as H is closed and yH is closed. Now, let V be a neighborhood of x such that $x \in V \subset G \setminus yH$.

Observe $Vx \cap yH = \emptyset$ implies $VxH \cap yH = \emptyset$ otherwise $vh_1 = gh_2$ and $vx = gh_2h_1^{-1}$ implying $Vx \cap yH \neq \emptyset$. Now, we can find a neighborhood of V_1 such that $V_1^{-1}V_1 \subset V$ therefore $V_1^{-1}V_1xH \cap yH = \emptyset$ this implies $V_1xH \cap V_1yH = \emptyset$, which is $\pi(V_1x) \cap \pi(V_1y) = \emptyset$ and by previous lemma $\pi(V_1x)$ and $\pi(V_1y)$ are open sets and $xH \in \pi(V_1x)$ and $yH \in \pi(V_1y)$. Hence □

$G \backslash H$ is T_2

Remark 0.3. We call $G \backslash H$ the quotient space where H is a closed subgroup of H, has equipped with quotient topology.

Lemma 0.3. *If G is a topological group, H is normal subgroup, the quotient topology equipped on the quotient group $G \backslash H$ is a topological group.*

Proof. We want to show $\chi_1 : G \backslash H \times G \backslash H \to G \backslash H$,

$$(x_1, y_1) \mapsto x_1 y_1^{-1}$$

is continuous.

And $\chi : G \times G \to G$ is continuous

$$(x, y) \mapsto xy^{-1}$$

□

$$\begin{array}{ccc} G \times G & \xrightarrow{\pi \times \pi} & G \backslash H \times G \backslash H \\ \downarrow \pi & & \downarrow \psi_1 \\ G & \xrightarrow{\psi} & G \backslash H \end{array}$$

Observe $\psi_1(\pi \times \pi) = \psi\pi$ and π, ψ are continuous. Hence $\psi_1(\pi \times \pi)$ are continuous.

Let O_1 be an open subset of $G\backslash H$ then $\psi_1(\pi \times \pi)^{-1}(O_1)$ is open. As $\pi \times \pi$ is an open map, $\pi \times \pi(\pi \times \pi)^{-1}\psi_1^{-1}(O_1)$ is open. Hence $\psi_1^{-1}(O_1)$ is open.

$\square$

Lemma 0.4. Let G be a topological group, and H be a subgroup (a) If G is compact then H and $G\backslash H$ are both compact. (b) If G is locally compact then H and $G\backslash H$ have both locally compact.

Proof. (a) H is a closed subset of a compact set and $G/H = \pi(G)$ where G is compact and π is continuous mapping. Hence a continuous map of compact set is compact.

(b) A closed subset of a locally compact set is locally compact.

Let S be locally compact, $T = \overline{S}$ and $T \subseteq S$. For $p \in T \cap S$, there exists a neighborhood of p in S, $\overline{U_p}$ st. $\overline{U_p}$ is compact. Observe for $p \in T$ there exists an open set $T \cap U_p$ in T such that $\overline{T \cap U_p} = \overline{T} \cap \overline{U_p}$ is compact as $\overline{T \cap U_p}$ is closed subset of $\overline{U_p}$ which is compact.

Hence H is locally compact.

Goal: G/H is locally compact.

$\square$

For $q_i \in G\backslash H$ and nbd of q_i. Let $U = \pi^{-1}(U_i)$, and let $x \in G$ st. $\pi(x) = q_i$.

As G is locally cpt we get a nbd of x $\overline{O}$ st. $\overline{O}$ is cpt and $\overline{O} \subseteq U$ by proposition given below. Then we have $\pi(\overline{O}) \subseteq \pi(U) = U_i$. If $O_i = \pi(O)$ a nbd of q_i. And $\pi(\overline{O})$ is a continuous image of cpt set hence we get $\pi(\overline{O})$ is cpt. Since G/H is T_2-space $\pi(\overline{O})$ is closed which implies

$O \subseteq \overline{O} \Rightarrow O_i \subseteq \pi(\overline{O})$ and $\overline{O_i} \subseteq \pi(\overline{O}) = \pi(\overline{O})$ which is cpt, so is $\overline{O_i}$. Thus O_i is a nbd of q_i such that $\overline{O_i}$ is cpt. It follows G/H is locally cpt.

Corollary

If G is locally cpt then G/H is normal or T_4.

Proposition 0.2. For any topological group G, let U be a neighborhood of e, there exists a neighborhood of e V such that $V \subseteq U$.

Proof. By property 5) and 6) we know that for every neighborhood of e U we have a symmetric neighborhood of e V such that $V^2 \subseteq U$. Now $x \in \overline{V} = \bigcap_{w \in V^e} w$ where V^e is set of all neighborhoods of e, this implies $x \in \overline{V} \subseteq V^2 \subseteq U$. $\square$